Is the straight-through estimator enough for quantized test-time adaptation?

A negative result on Gaussian-smoothed surrogate gradients. ResNet-20, CIFAR-10 / CIFAR-10-C, W8A8 / W4A4 / W3A3.

1. Question

Gradient-based test-time adaptation (TTA) of a quantized network must backpropagate through a staircase quantizer whose true derivative is zero almost everywhere. The standard fix is the straight-through estimator (STE). We asked whether replacing the quantizer's backward pass with a **Gaussian-smoothed surrogate gradient**, optionally with a **label-free adaptive smoothing width**, improves QTTA over a well-tuned STE. The forward pass is unchanged.

2. Setup

- **Model:** CIFAR ResNet-20 (about 0.27M parameters), trained in FP32 to 92.79% clean accuracy.
- **Quantization:** post-training, per-channel MSE-clipped weights, per-tensor MSE-clipped input activations, first conv and final FC at 8 bits, BatchNorm kept separate (not folded). Clean accuracy: W8A8 about 92.8%, W4A4 about 90.0%, W3A3 about 57.6%.
- **Surrogate gradient:** for a uniform quantizer, the smoothed derivative is a sum of unit-area Gaussian bumps at the rounding thresholds (zero in the saturated region). Implemented as a custom autograd function and checked against the exact smoothed quantizer (max error below 5e-4 for sigma/Delta from 0.1 to 2). Forward outputs are bit-identical to the plain quantizer.
- **TTA protocol:** TENT-style entropy minimization on BatchNorm affine parameters, episodic per corruption, online within it, batch size 64, Adam (lr 1e-3), CIFAR-10-C severity 5. Accuracy is measured on the prediction made before each update.
- **Controls:** no TTA, BN-statistics-only adaptation, FP32 and W8A8 ceilings, harder quantizers (W3A3, min-max W4A4), optimizer stress (lr, batch size), adapting weights as well, and a held-out oracle for the best possible corruption-dependent sigma.

3. Results

3.1 Narrow smoothing hurts; wide smoothing becomes STE. At sigma/Delta = 0.1 the surrogate is spiky (gradient norm 27x STE, cosine similarity to STE of -0.08). For sigma/Delta of 0.5 or more it is nearly STE (cosine 0.92 to 0.99). In the 5-corruption sweep at W4A4, STE reached 75.46% mean accuracy; sigma/Delta = 0.15 reached 72.76%, 0.25 reached 74.94%, and 1.0 reached 75.64% (+0.18, about one seed standard deviation).

3.2 STE already matches exact gradients for BN-only adaptation. The gain of entropy minimization over BN-statistics-only adaptation is 4.23 points for FP32 (no quantizer, exact gradients) and 4.32, 4.36, 4.40, 4.54 points for W8A8, W4A4, W4A4-minmax and W3A3. The remaining gap to FP32 (1.9 points at W4A4, 5.1 for min-max, 7.2 at W3A3) is already present with BN statistics only, so it comes from quantization error in the forward pass, not from gradient quality.

3.3 No headroom for adaptive sigma. Over all 15 standard corruptions and 3 seeds, a per-corruption oracle that picks sigma using labels was evaluated with leave-one-seed-out selection to remove winner's-curse optimism. Relative to the best single fixed sigma it gains 0.00 points at W4A4 (95% CI -0.07 to +0.09) and -0.02 at W3A3 (CI -0.08 to +0.05). The best fixed sigma beats STE by +0.11 at W4A4 (CI -0.00 to +0.21) and -0.00 at W3A3 (CI -0.11 to +0.11). The pre-registered criterion for building an adaptive rule (CI lower bound above 0.3 points) was not met by a wide margin.

3.4 Stress settings. With lr 1e-2 or batch size 16, results are mixed in sign. Gaussian smoothing is clearly worse at lr 1e-2 with batch 64 (-0.4 to -8.6 points), where STE is itself degraded, and every method collapses at lr 1e-2 with batch 16.

3.5 Adapting weights too. Updating conv weights through the weight quantizer gives no gain over BN-only (STE: 74.54 vs 74.55 at W4A4; 69.19 vs 69.94 at W3A3 for lr_w 1e-4) and collapses at lr_w 5e-4. Sigma/Delta = 0.25 beats STE there (+2 to +4 points) but only inside the collapsed regime (at most 64%, versus 74.5% for BN-only).

3.6 Cost. The exact Gaussian backward pass was 2.5 to 4 times slower per run than STE. For sigma/Delta of 1 or more, a closed-form envelope (two erf calls) approximates it within about 1e-2 at the clip edges; its speed-up was not measured.

4. Conclusion

For BN-only quantized TTA on ResNet-20, STE is sufficient: it delivers the same adaptation gain as exact gradients, Gaussian smoothing at best matches it within noise, and a label-aware choice of sigma per corruption adds nothing measurable. The remaining accuracy

loss is forward quantization error, which a better backward surrogate cannot fix.

5. Limitations

- One small architecture and dataset; deeper or larger models, other bit-widths and other shifts may behave differently.
- One TTA objective (entropy) and mostly BN-only updates; learnable step sizes (LSQ-style scale gradients) were not tested.
- Seeds change only batch order, so uncertainty is estimated mainly across corruptions; the stress scans use 2 seeds and 3 corruptions.
- Only a clipped-STE baseline and a Gaussian surrogate were compared; other estimators were not.
- An FP32 weight-adaptation control was not run, so it is unknown whether the weight-adaptation collapse is caused by STE or by entropy minimization itself.
- No systematic literature search was done, so novelty of any of these components is unassessed.

6. Reproducibility

Code: `qtta_lib.py` (model, quantizers, surrogate gradient) and notebooks 01 to 07 (training, tests, TTA sweeps, oracle test, figures). Figures: sigma curve, gradient similarity and ceiling, stress scan, weight adaptation, oracle headroom.